Interpreter decision guide

Operational criteria for crop-system, crop-type, and crop-rotation interpretation in the Brazilian Cerrado

Supplementary Document S1 | Cerrado crop labels, dataset v1.0

1. Purpose and scope

This guide records the operational criteria that interpreters applied in Collect Earth Online (CEO) when assigning annual crop-system, crop-type, and crop-rotation labels to the 170 sample locations of the study. Each label refers to one sample location in one August–July agricultural year, from August 2001–July 2002 to August 2024–July 2025. The criteria are summarized graphically in Figures 3 and 4 of the accompanying article.

The rules are interpretation guidance, not automated classification thresholds. Interpreters applied them together with contextual evidence from Landsat composites and high-resolution imagery, and recorded uncertain cases for group review. Labels produced with this guide are remotely interpreted; they are not independently validated field observations.

2. Evidence sources

Source	Use in interpretation	Repository resource
MODIS EVI time series (MOD13Q1, 16-day composites, 250 m)	Phenology: peak magnitude, number of peaks, peak timing, peak width, persistence, and curve shape	gee/modis.js
Landsat seasonal composites	Seasonal colour, spatial texture, field geometry, and evidence of conversion or a second crop	gee/landsat.js
High-resolution basemaps in CEO	Field boundaries, mixed-pixel screening, and irrigation infrastructure such as central pivots	Viewed within CEO

EVI scaling. MODIS EVI thresholds in this guide are raw MOD13Q1 integer values. Multiply by 0.0001 to obtain EVI on the 0–1 scale; for example, 5,000 corresponds to an EVI of 0.50.

3. Interpretation sequence

For each point-year, interpreters worked through the following steps:

1. Screen the location for cropland presence and for mixed or edge pixels (Section 4).
2. Assign the crop-system class: non-cropped, single cropped, double cropped (non-irrigated), irrigated crop, or other (Section 5).
3. Where the evidence was sufficient, assign the crop type or crop rotation (Section 6).
4. Record cover-crop status.
5. Record uncertainty, imagery limitations, or possible relocation in the notes field (Section 7).

4. Cropland presence and conversion

- Cropland was confirmed using geometric and texture clues (rectangular or circular fields, uniform texture and colour), together with clear changes in Landsat mosaics indicating conversion from pasture or natural vegetation to row crops.
- Points that fell on field edges, roads, or mixed pixels were marked as poor quality and flagged for review. After review, such points were either relocated to a homogeneous field interior or excluded under a documented quality-control decision.

5. MODIS EVI thresholds and phenology

Pattern	Raw MOD13Q1 value (EVI, 0–1 scale)	Interpretation
Peak EVI above threshold	> 5,000 (> 0.50)	Candidate for cropped land (higher-productivity crops).
Intermediate peak EVI	3,000–5,000 (0.30–0.50)	Generally not classified as cropped (lower confidence).
True double peak	> 5,000 (> 0.50) in two distinct growing periods	Double cropping, where each peak spans about 5 or more 16-day observations and Landsat imagery shows evidence of a second crop.
Dry-season persistence	> 6,000 (> 0.60) through June–August	Irrigated for that agricultural year; also assigned when central-pivot infrastructure is visible in Landsat or basemaps.

6. Crop-type and crop-rotation assignment

Crop types were assigned hierarchically and cross-checked against Landsat timing and texture.

- **Soy (first crop).** Growth starting in September–December and peaking in December–February indicates a likely soy first (wet-season) crop.
- **Corn (second crop).** In non-irrigated double cropping, a second crop in March–May is likely corn if Landsat shows red or dark-orange tones with spatial heterogeneity during March–May.
- **Cotton (second crop).** A second crop in March–May is likely cotton if Landsat shows lighter-orange tones with very uniform texture.
- **Sugarcane.** Long-duration high EVI (about nine months near peak greenness) with a jagged, highly variable plateau lasting more than six months, together with distinctive Landsat texture or terracing, indicates sugarcane.
- **Cover crop or alternative rotation.** A small early peak followed by a large January–July peak may indicate a cover crop followed by cotton, or another rotation. Interpreters resolved these cases using both Landsat colour and EVI timing.
- **Irrigated crops.** Where central-pivot or other irrigation features were observed, the label was recorded as irrigated together with the crop type.

Standardized classes

Recorded responses were harmonized into the standardized classes used in the dataset. The full mapping from CEO response values to standardized classes is provided in data/crop_class_mapping.csv (Supplementary Table S1).

Class type	Standardized classes
Single crop	Soy, Corn, Cotton, Sugarcane, Rice, Other
Double-crop rotation	Soy–corn, Soy–cotton, Soy–other
Auxiliary (retained in the audit dataset)	Non-cropped, Irrigated

7. Uncertainty, review, and adjudication

- Interpreters recorded uncertainty, possible mixed-pixel conditions, potential point relocation, imagery limitations, and classes requiring review in the CEO notes field.
- Where the available imagery and EVI patterns were insufficient to assign a crop-system or crop-rotation label under this guide, no crop label was assigned.
- Flagged cases and interpreter disagreements were reviewed by the interpretation team. Where an adjudication decision was made, interpreters updated their CEO entries to reflect the agreed interpretation.
- Notes informed quality-control review but did not by themselves exclude a record. Exclusion required an explicit location-level or year-specific QC decision, documented in the dataset.

8. Limitations

- The thresholds were developed for MODIS MOD13Q1 EVI and Cerrado cropping calendars. They should be revised with local expert input before use in other regions or with other sensors.
- At 250 m, MODIS pixels can mix several land covers, particularly near field edges and in smaller fields; Landsat and high-resolution imagery were used to screen these cases.
- Cover crops, second crops, and alternative rotations can produce similar EVI patterns. These cases relied on interpreter judgement using both Landsat colour and EVI timing.
- Applying this guide does not establish independent label accuracy.